

Mahrukh

Report: Design, Implementation, and Testing of a Custom Firewall

Objective

The objective of this lab was to design, implement, and test a custom open-source firewall system similar to pfSense using Kali Linux, with validation performed using a Windows system. The firewall demonstrates packet filtering, stateful firewall rules, Network Address Translation (NAT), logging & monitoring, and basic web-based firewall management.

Network Topology and Adapter Configuration

- **Kali Linux (Firewall):**
 - Adapter 1: NAT (WAN interface)
 - Adapter 2: Host-Only (LAN interface)
- **Windows Machine (Client):**
 - Adapter: Host-Only

This topology enabled internal communication between Windows and Kali through the Host-Only network, while NAT allowed Kali to act as a gateway.

Deployment on Virtual Machines

The firewall was deployed using virtual machines. Kali Linux acted as the firewall system, while Windows was used as a client machine. This setup allowed controlled testing of firewall behavior in an isolated virtual environment.

Firewall Implementation and Execution

Bash-Based Firewall Setup

The firewall was configured using bash scripts executed directly on the Kali Linux interface. These scripts were used to set default policies, apply firewall rules, configure NAT, and enable logging. Execution of the bash scripts initialized and activated the firewall on Kali.

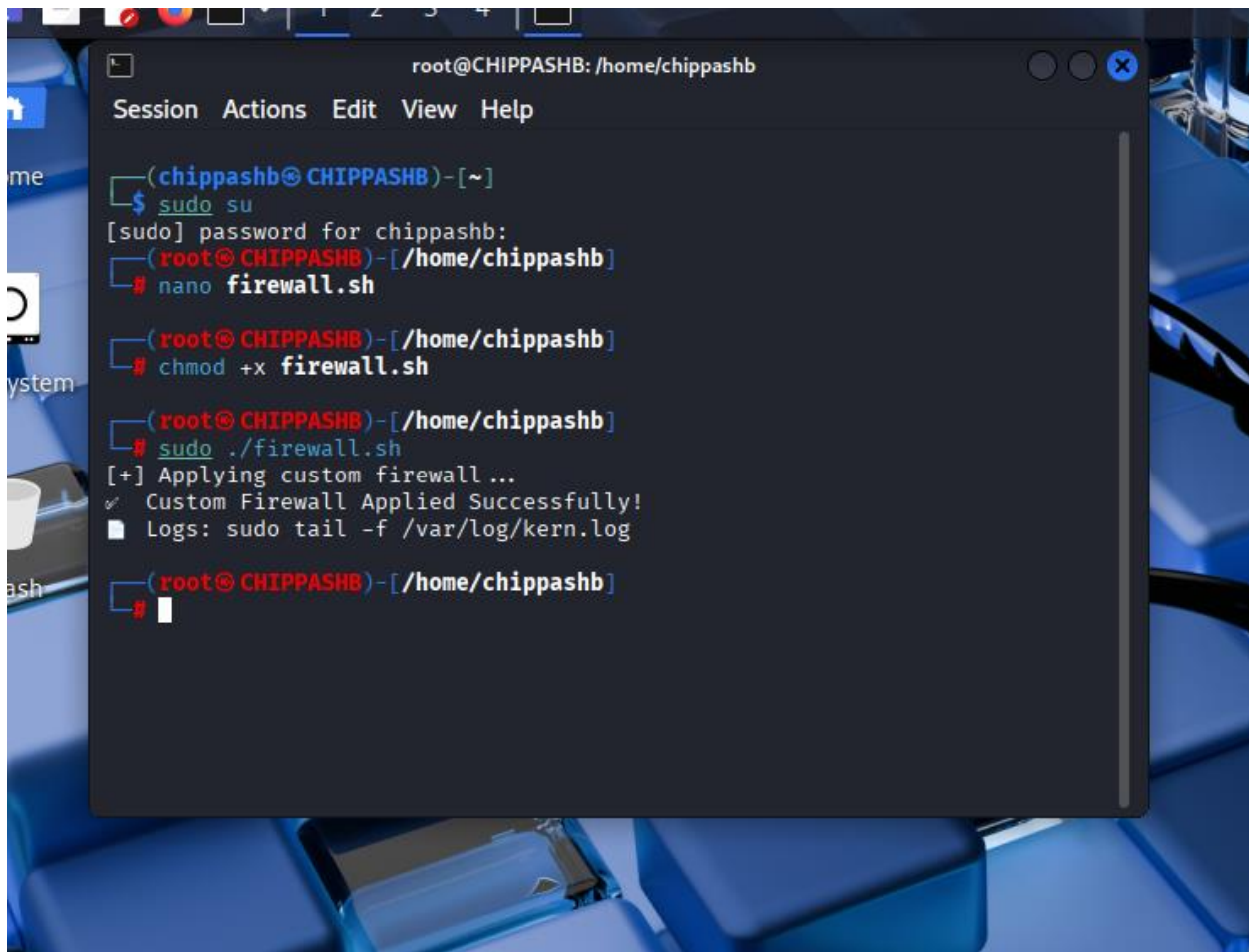
```
chippashb@CHIPPASHB: ~
Session Actions Edit View Help
GNU nano 8.6 FIREWALL.sh *
echo 1 > /proc/sys/net/ipv4/ip_forward

# NAT for LAN → WAN
iptables -t nat -A POSTROUTING -s $LAN_SUBNET -o $WAN_IF -j MASQUERADE

# Forward traffic
iptables -A FORWARD -i $LAN_IF -o $WAN_IF -j ACCEPT
iptables -A FORWARD -i $WAN_IF -o $LAN_IF -m state --state ESTABLISHED,RE>

#
# 8. Logging (limited)
#
iptables -A INPUT -m limit --limit 5/min -j LOG \
--log-prefix "FW-INPUT-DROP: " --log-level 4
iptables -A FORWARD -m limit --limit 5/min -j LOG \
--log-prefix "FW-FORWARD-DROP: " --log-level 4

#
# 9. Explicit drop (last rule)
#
Save modified buffer?
Y Yes
N No ^C Cancel
```

A terminal window titled 'root@CHIPPASHB: /home/chippashb' with a menu bar (Session, Actions, Edit, View, Help). The terminal shows a user 'chippashb' at 'CHIPPASHB' in their home directory. They run 'sudo su' to become root. The prompt changes to 'root@CHIPPASHB' in '/home/chippashb'. They run 'nano firewall.sh' to create a script. Then 'chmod +x firewall.sh' to make it executable. Finally, they run 'sudo ./firewall.sh'. The script outputs: '[+] Applying custom firewall ...', '✓ Custom Firewall Applied Successfully!', and '■ Logs: sudo tail -f /var/log/kern.log'. The prompt returns to the user 'chippashb' in their home directory.

```
root@CHIPPASHB: /home/chippashb
Session Actions Edit View Help

(chippashb@CHIPPASHB)-[~]
$ sudo su
[sudo] password for chippashb:
(root@CHIPPASHB)-[/home/chippashb]
# nano firewall.sh

(root@CHIPPASHB)-[/home/chippashb]
# chmod +x firewall.sh

(root@CHIPPASHB)-[/home/chippashb]
# sudo ./firewall.sh
[+] Applying custom firewall ...
✓ Custom Firewall Applied Successfully!
■ Logs: sudo tail -f /var/log/kern.log

(chippashb@CHIPPASHB)-[/home/chippashb]
#
```

Python-Based Web Integration

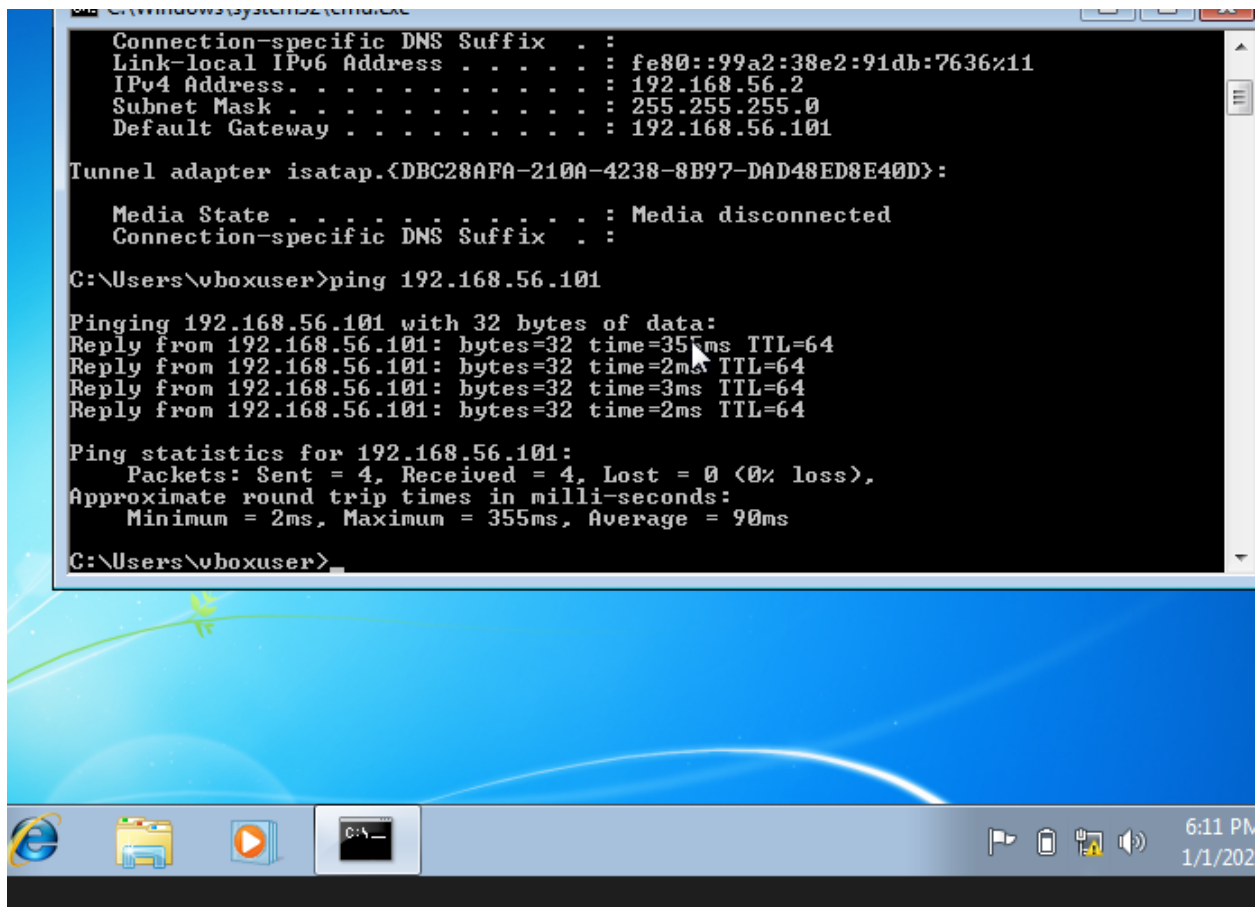
Python scripts were executed on Kali to support firewall monitoring and web-based management. The Python component connects with the firewall by reading firewall status and rule information, which is then displayed through a web interface. This integration allows firewall behavior configured via bash to be monitored through the web interface.

A web-based management interface was implemented using Python. The interface provides a simple browser-based view of firewall activity and status, reducing reliance on command-line monitoring and improving manageability.

Testing and Validation:

1. Windows and Kali Connectivity Testing

Connectivity between Windows and Kali was first verified using their assigned IP addresses. Successful communication confirmed correct adapter configuration and network setup.



```
C:\Windows\system32\cmd.exe
Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::99a2:38e2:91db:7636%11
IPv4 Address. . . . . : 192.168.56.2
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.56.101

Tunnel adapter isatap.{DBC28AFA-210A-4238-8B97-DAD48ED8E40D}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

C:\Users\vboxuser>ping 192.168.56.101

Pinging 192.168.56.101 with 32 bytes of data:
Reply from 192.168.56.101: bytes=32 time=355ms TTL=64
Reply from 192.168.56.101: bytes=32 time=2ms TTL=64
Reply from 192.168.56.101: bytes=32 time=3ms TTL=64
Reply from 192.168.56.101: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.56.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 355ms, Average = 90ms

C:\Users\vboxuser>
```

Custom Firewall Manager

Add Packet Filtering Rule

Protocol:

Source IP:

Destination IP:

Source Port:

Destination Port:

Action:

DROP

Add Rule

Enable/Disable NAT

Enable NAT

Disable NAT

Chain	Rule	Action
1	ACCEPT all -- 0.0.0.0/0 0.0.0.0/0 state RELATED,ESTABLISHED	Delete
2	LOG all -- 0.0.0.0/0 0.0.0.0/0 LOG flags 0 level 4 prefix "Firewall: "	Delete

```
Logs
Jan 01 08:19:28 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=160 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:19:41 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=170 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:19:42 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=171 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:19:50 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=182 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:19:56 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=183 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:19:57 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=184 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:20:10 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=194 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:20:11 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=195 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:20:11 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=196 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:20:25 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=206 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:20:26 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=207 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:20:26 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=208 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:20:40 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=218 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:20:40 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=219 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:20:41 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=220 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:20:42 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=223 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:20:43 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=224 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:20:44 CHIPPASH kernel: Firewall: INeth1 OUT= MAC=ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=225 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:24:40 CHIPPASH kernel: Firewall: INlo OUT= MAC=00:00:00:00:00:00:00:00:00:00:00:00 SRC=127.0.0.1 DST=127.0.0.1 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=14685 DF PROTO=TCP SPT=56532 DPT=5800 WINDOW=65535 RES=0x00 SYN URG=0
```

Monitoring (Packet Counts)

Chain INPUT (policy ACCEPT 53 packets, 4320 bytes)					destination		state RELATED,ESTABLISHED	
pkts	bytes	target	prot	opt	in	out	source	
4	624	ACCEPT	all	..	*	*	0.0.0.0/0	
1	60	LOG	all	..	*	*	0.0.0.0/0	LOG flags 0 level 4 prefix "Firewall: "
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)					destination			
pkts	bytes	target	prot	opt	in	out	source	
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)					destination			
pkts	bytes	target	prot	opt	in	out	source	

2. Traffic Blocking and Web Control Testing

After validation, firewall rules were applied to block Windows traffic. Access that was initially allowed was denied, and the effect of blocking was observed through the firewall behavior and web-based management interface.

Blocking the IP:

Custom Firewall Manager

Add Packet Filtering Rule

Protocol:

Source IP:

Destination IP:

Source Port:

Destination Port:

Action:

Enable/Disable NAT

OUTPUT:

```
C:\Users\vboxuser>ping 192.168.56.101
Pinging 192.168.56.101 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.56.101:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\vboxuser>
```

Enable/Disable NAT

Enable NAT

Disable NAT

Current Rules

Chain	Rule	Action
1	ACCEPT all -- 0.0.0.0/0 0.0.0.0/0 state RELATED,ESTABLISHED	Delete
2	LOG all -- 0.0.0.0/0 0.0.0.0/0 LOG flags 0 level 4 prefix "Firewall: "	Delete
3	DROP icmp -- 192.168.56.2 192.168.56.101	Delete

Logs

```
Jan 01 00:50:50 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=467 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:00:11 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=477 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:00:12 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=478 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:00:12 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=479 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:00:26 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=489 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:00:26 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=490 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:00:27 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=491 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:00:40 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=501 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:00:41 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=502 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:00:42 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=503 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:00:51 CHIPPASHB kernel: Firewall: 1InetO OUT= MAC=00:00:00:00:00:00:00:00:00:00:00:00 SRC=127.0.0.1 DST=127.0.0.1 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=11436 DF PROTO=TCP SPT=40384 DPT=5000 WINDOW=65495 RES=0x00 SYN URG=0
Jan 01 00:00:52 CHIPPASHB kernel: Firewall: 1InetO OUT= MAC=00:00:00:00:00:00:00:00:00:00:00:00 SRC=127.0.0.1 DST=127.0.0.1 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=37881 DF PROTO=TCP SPT=40388 DPT=5000 WINDOW=65495 RES=0x00 SYN URG=0
Jan 01 00:00:55 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=513 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:00:55 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=514 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:00:56 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=515 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:01:09 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=525 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:01:10 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=526 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:01:11 CHIPPASHB kernel: Firewall: 1Ineth1 OUT= MAC=ffff:ffff:ffff:ffff:08:00:27:13:80:8c:00:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TOS=0x00 PREC=0x00 TTL=128 ID=527 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 00:01:17 CHIPPASHB kernel: Firewall: 1InetO OUT= MAC=00:00:00:00:00:00:00:00:00:00:00:00 SRC=127.0.0.1 DST=127.0.0.1 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=11546 DF PROTO=TCP SPT=50450 DPT=5000 WINDOW=65495 RES=0x00 SYN URG=0
Jan 01 00:01:17 CHIPPASHB kernel: Firewall: 1InetO OUT= MAC=00:00:00:00:00:00:00:00:00:00:00:00 SRC=127.0.0.1 DST=127.0.0.1 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=31515 DF PROTO=TCP SPT=50450 DPT=5000 WINDOW=65495 RES=0x00 SYN URG=0
```

Monitoring (Packet Counts)

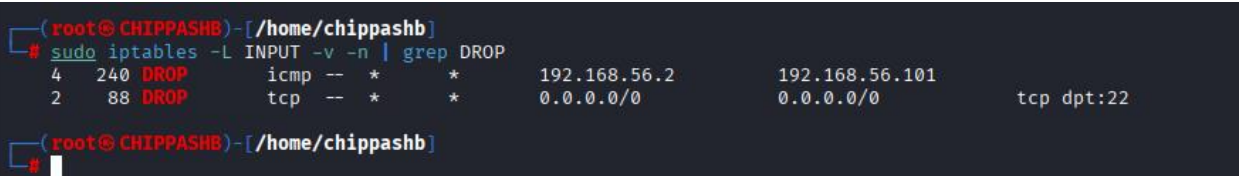
```
Chain INPUT (policy ACCEPT 148 packets, 11351 bytes)
pkts bytes target prot opt in out source destination
33 11284 ACCEPT all -- * * 0.0.0.0/0 0.0.0.0/0 state RELATED,ESTABLISHED
10 780 LOG all -- * * 0.0.0.0/0 0.0.0.0/0 LOG flags 0 level 4 prefix "Firewall: "
0 0 DROP icmp -- * * 192.168.56.2 192.168.56.101

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target prot opt in out source destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target prot opt in out source destination
```

Packet Filtering

Packet filtering was enforced using default deny policies. Only explicitly allowed traffic was permitted, while unauthorized packets were blocked, ensuring controlled access between Windows and Kali.



Stateful Firewall Rules

Stateful firewall rules were applied to allow return traffic only for established and related connections. This ensured that legitimate responses were allowed while unsolicited traffic was denied.


```
(root@CHIPPASHB)-[/home/chippashb]
# sudo iptables -L INPUT -v -n
Chain INPUT (policy ACCEPT 4 packets, 409 bytes)
pkts bytes target      prot opt in      out     source        destin
ation
23  6285 ACCEPT      all  --  *      *       0.0.0.0/0     0.0.0.
0/0
state RELATED,ESTABLISHED
4   409 LOG        all  --  *      *       0.0.0.0/0     0.0.0.
0/0
LOG flags 0 level 4 prefix "Firewall: "

(root@CHIPPASHB)-[/home/chippashb]
#
```

Network Address Translation (NAT)

NAT was configured on the Kali firewall to translate internal traffic to the external interface. This allowed internal hosts to communicate externally while hiding internal IP addresses.

```
(root@CHIPPASHB)-[/home/chippashb]
# sudo iptables -t nat -L -v -n
Chain PREROUTING (policy ACCEPT 5 packets, 355 bytes)
pkts bytes target      prot opt in      out     source        destination
0     0 DNAT        tcp  --  *      *       0.0.0.0/0     0.0.0.0/0
5   355 LOG        all  --  *      *       0.0.0.0/0     0.0.0.0/0
tcp dpt:80 to:192.168.56.101:8080
LOG flags 0 level 4 prefix "NAT-Test: "

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in      out     source        destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in      out     source        destination

Chain POSTROUTING (policy ACCEPT 4 packets, 718 bytes)
pkts bytes target      prot opt in      out     source        destination
0     0 MASQUERADE  all  --  *      eth0    0.0.0.0/0     0.0.0.0/0
```

Logging and Monitoring

Logging and monitoring were enabled to track blocked and unauthorized traffic. Firewall activity was monitored through logs and the web-based interface, confirming active enforcement of security rules.

```
(root@CHIPPASHB)-[/home/chippashb]
# sudo journalctl -k | grep Firewall
Jan 01 08:03:00 CHIPPASHB kernel: Firewall: IN=lo OUT= MAC=00:00:00:00:00:00:
00:00:00:00:00:00:00:08:00 SRC=127.0.0.1 DST=127.0.0.1 LEN=60 TOS=0x00 PRE
C=0x00 TTL=64 ID=50483 DF PROTO=TCP SPT=45914 DPT=5000 WINDOW=65495 RES=0x
00 SYN URG=0
Jan 01 08:03:01 CHIPPASHB kernel: Firewall: IN=lo OUT= MAC=00:00:00:00:00:00:
00:00:00:00:00:00:00:08:00 SRC=127.0.0.1 DST=127.0.0.1 LEN=60 TOS=0x00 PRE
C=0x00 TTL=64 ID=40481 DF PROTO=TCP SPT=45916 DPT=5000 WINDOW=65495 RES=0x
00 SYN URG=0
Jan 01 08:07:21 CHIPPASHB kernel: Firewall: IN=eth1 OUT= MAC=ff:ff:ff:ff:f
f:ff:0a:00:27:00:00:13:08:00 SRC=192.168.56.1 DST=192.168.56.255 LEN=229 T
OS=0x00 PREC=0x00 TTL=128 ID=43276 PROTO=UDP SPT=138 DPT=138 LEN=209
Jan 01 08:11:06 CHIPPASHB kernel: Firewall: IN=eth1 OUT= MAC=08:00:27:16:6
4:48:08:00:27:13:80:8c:08:00 SRC=192.168.56.2 DST=192.168.56.101 LEN=60 TO
S=0x00 PREC=0x00 TTL=128 ID=52 PROTO=ICMP TYPE=8 CODE=0 ID=1 SEQ=1
Jan 01 08:17:15 CHIPPASHB kernel: Firewall: IN=lo OUT= MAC=00:00:00:00:00:00:
00:00:00:00:00:00:00:08:00 SRC=127.0.0.1 DST=127.0.0.1 LEN=60 TOS=0x00 PRE
C=0x00 TTL=64 ID=5008 DF PROTO=TCP SPT=56128 DPT=5000 WINDOW=65495 RES=0x0
0 SYN URG=0
Jan 01 08:17:29 CHIPPASHB kernel: Firewall: IN=eth1 OUT= MAC=ff:ff:ff:ff:f
f:ff:08:00:27:13:80:8c:08:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TO
S=0x00 PREC=0x00 TTL=128 ID=62 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:17:30 CHIPPASHB kernel: Firewall: IN=eth1 OUT= MAC=ff:ff:ff:ff:f
f:ff:08:00:27:13:80:8c:08:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TO
S=0x00 PREC=0x00 TTL=128 ID=63 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:17:31 CHIPPASHB kernel: Firewall: IN=eth1 OUT= MAC=ff:ff:ff:ff:f
f:ff:08:00:27:13:80:8c:08:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TO
S=0x00 PREC=0x00 TTL=128 ID=64 PROTO=UDP SPT=137 DPT=137 LEN=58
Jan 01 08:17:44 CHIPPASHB kernel: Firewall: IN=eth1 OUT= MAC=ff:ff:ff:ff:f
f:ff:08:00:27:13:80:8c:08:00 SRC=192.168.56.2 DST=192.168.56.255 LEN=78 TO
S=0x00 PREC=0x00 TTL=128 ID=74 PROTO=UDP SPT=137 DPT=137 LEN=58
```

Conclusion

The firewall was successfully deployed and tested on virtual machines. Web-based management, packet filtering, stateful rules, NAT, and logging were all implemented and validated. The system demonstrates core functionality comparable to a basic pfSense-style firewall.

The custom firewall system was successfully implemented and tested using Kali Linux and Windows. Bash-based execution ensured correct firewall enforcement, while Python-based web management improved visibility. Connectivity testing followed by controlled blocking verified correct firewall behavior, demonstrating functionality comparable to a basic pfSense-style firewall system.